Omninet AI Usability Test Plan

1. Background and Purpose

Omninet AI is a self-service Internet kiosk deployed in public places (airports, shopping malls, hotels, etc.). Users interact with the kiosk via a touchscreen to make payments, browse the web, and use an embedded AI Assistant for queries and translation. The system also includes a Support Center for operators to monitor and manage the kiosk network.

Test Objectives:

- Evaluate the **effectiveness, efficiency, and satisfaction** of target users when performing real tasks with the Omninet AI kiosk.
- Identify usability problems (especially for critical paths: AI assistant interaction, payment flow, language selection, etc.).
- Collect quantitative and qualitative data to provide actionable recommendations for product iteration.

2. Test Scope

This usability test focuses on the **Kiosk** side, covering core user tasks:

- Start session and pay (cash, credit card, mobile wallet)
- Browse the web and use the AI Assistant (query, translation, summarization)
- Purchase additional time and terminate session
- Multi-language selection and localization adaptation

Usability of the Support Center will be covered in a separate test.

3. User Personas and Participants

Based on the target usage scenarios defined in the MRD, recruit **8–12 representative participants**in the following categories:

User Type	Characteristics
Traveler	First-time use at airports/stations, time-pressured, may not be fluent in local language

User Type	Characteristics
Shopper	Leisure scenario, willing to try new features (e.g., AI Assistant), longer dwell time
Low-tech literacy	Unfamiliar with touchscreens and digital payments, needs simple, intuitive guidance

Recruitment: Online or on-site screening. Ensure diversity in age, gender, and technical experience. Participants should not have used Omninet AI before (to preserve learnability measurement).

4. Test Objectives and Metrics

Based on ISO 9241-11 and Nielsen's model, define the following objectives and **quantitative metrics**:

Objective	Metric	Minimum Acceptable	Target
Effectiveness	Task completion rate (%)	⩾ 85%	⩾ 95%
Efficiency	Average task completion time (seconds)	Pay and start browsing ⩽ 90 sec	⩽ 60 sec
Efficiency	Number of steps (touches) per task	⩽ 8	⩽ 5
Error rate	Number of user errors per task (needing correction)	⩽ 2	⩽ 1
Learnability	Time reduction between first and second use of same task	⩾ 30%	⩾ 50%
Satisfaction	System Usability Scale (SUS) score	⩾ 70	⩾ 80

Objective	Metric	Minimum Acceptable	Target
AI Assistant satisfaction	Likert scale (1–5) on helpfulness, speed, naturalness	≥ 4.0 avg	≥ 4.5 avg

5. Test Tasks (Scenarios)

Design **5 core tasks**, each representing a real-world scenario. Task descriptions are written in user language, avoiding system jargon.

Task 1: Start session and pay with cash

- **Scenario:** You are at the airport with 30 minutes of free time. You want to check email. You have $10 in cash.
- **Task:** Start the kiosk, select English, pay $5 in cash for 30 minutes of Internet access, and reach the browser home page.

Task 2: Use the AI Assistant to get information

- **Scenario:** While browsing local tourism information, you want to find Chinese restaurants nearby.
- **Task:** Open any webpage, click the AI Assistant icon, type or speak "Recommend nearby Chinese restaurants", and read the answer.

Task 3: Purchase additional time

- **Scenario:** Your remaining time is 1 minute, and a pop-up reminds you.
- **Task:** Use a credit card (test card) to buy another 15 minutes and successfully extend the session.

Task 4: Language switching and translation

- **Scenario:** You are a Spanish-speaking user and the kiosk defaults to English.

- **Task:** Switch the language to Spanish on the welcome screen, then browse an English news website, and use the AI Assistant to translate a paragraph into Spanish.

Task 5: End the session

- **Task:** Actively click the "Log out" button, confirm session termination, and observe the browser returning to the welcome screen.

6. Test Method

Empirical approach using **Think Aloud** and **observation recording**.

- **Type:** Formative evaluation – to discover design problems and improve.
- **Environment:** Lab simulation with Omninet AI kiosk prototype (or high-fidelity clickable prototype).
- **Data collection:**
    - Screen recording + camera capturing facial expressions
    - Test log (manual recording of time, errors, observations)
    - Post-task questionnaires (SUS + custom AI Assistant questionnaire)
    - Final semi-structured interview

Procedure (approx. 60 minutes per participant):

1. **Welcome & informed consent** (5 min)
2. **Background questionnaire** (5 min)
3. **Task instructions** – "We are testing the system, not you"; explain think-aloud (5 min)
4. **Perform 5 tasks** (30 min, ~6 min per task)
5. **Post-task questionnaires (SUS + custom)** (10 min)
6. **Brief interview** – overall impression, hardest / most liked parts (5 min)

7. Test Environment and Equipment

Equipment	Specification
Kiosk hardware	24-inch touchscreen all-in-one running Omninet AI simulation (or high-fidelity Figma prototype with interaction)
Network	Simulated 5G/WiFi environment

Equipment	Specification
Payment simulation	Test card reader or digital wallet sandbox
Recording equipment	Camera (face), screen recording software (e.g., Morae, OBS)
Timer	Stopwatch or automatic event logging

8. Data Collection and Analysis

8.1 Quantitative Data

- Success/failure (binary) per task
- Task completion time (seconds)
- Number of steps (touch count)
- Error types and counts (e.g., wrong button, abandoned payment, language option not found)
- SUS total score (0–100)
- Average Likert score for AI Assistant

8.2 Qualitative Data

- Think-aloud comments (transcribed and coded)
- Observed confusion, hesitation, facial expressions
- Improvement suggestions from final interview

8.3 Analysis Methods

- Calculate task completion rate, average time, standard deviation.
- Descriptive statistics for SUS scores.
- Classify errors by severity (critical / non-critical).
- Use affinity diagram to group qualitative feedback and identify recurring issues.

9. Reporting and Recommendations

9.1 Report Structure

- **Executive summary** – key findings and recommendations
- **Methodology** – participants, tasks, environment
- **Results:**
  - Effectiveness / efficiency / satisfaction tables
  - Task completion time chart
  - SUS radar plot
  - Critical error list
- **Qualitative findings** – user quotes, typical pain points
- **Recommendations** – prioritised (high/medium/low)

9.2 Deliverables (examples)

- Structured Data Summary – per user, per task completion, time, errors
- Curve diagram – average task times with individual scatter
- Video clips – illustrating most critical usability problems (e.g., language switch not found)
- PowerPoint summary – for product owners and developers

9.3 Expected Improvements (illustrative)

- **High priority:** Language selection must show both native script and current language (already required in MRD v1.3).
- **High priority:** AI Assistant response time < 2 seconds, otherwise users will click repeatedly.
- **Medium priority:** Cash payment must clearly display accepted denominations and change policy.
- **Low priority:** Increase visual prominence of the "Log out" button (currently missed by some users).

10. Ethical Considerations

- All participants sign informed consent, with clear explanation of video recording and data use.
- Data is anonymised; reports contain no personal identification.
- Participants may withdraw at any time without penalty.
- If a participant shows obvious frustration, the test leader may pause and offer reassurance.

This plan is ready to be used for usability testing of the Omninet AI kiosk. It can be iterated as the project evolves (e.g., early paper prototypes, later full hardware).

Omninet AI Usability Test Report (Simulated)

1. Executive Summary

Product tested: Omninet AI public kiosk (Kiosk side)
Test dates: April 15 – April 18, 2026
Participants: 10 representative users (travelers, shoppers, low-tech literacy users)
Key findings:

- **Overall usability acceptable:** Average SUS score **76.5** (above minimum acceptable of 70).
- **Task completion rate:** Overall **88%** – below the ideal target of 95%, with major blockers in language switching and AI Assistant translation.
- **Efficiency issues:** Average time to pay and start browsing was **78 seconds** (above target of 60 seconds), mainly due to unclear guidance for cash payment change.
- **AI Assistant:** Users rated its **usefulness high (4.3/5)**, but response speed was slow in some tasks (> 3 seconds).
- **Most severe problem:** Language selection screen became unusable in some language modes (e.g., after switching to Hebrew, users could not find their way back to English).

Top recommendations:

1. Show language options in both native script and current interface language on the welcome screen.
2. Improve cash payment flow – clearly display accepted denominations and change handling.
3. Improve AI Assistant response time to under 2 seconds, add loading animation.
4. Make the "Log out" button more visually prominent.

2. Methodology

2.1 Participants

A total of **10 participants** were recruited:

User Type	Number	Average Age	Self-rated Tech Experience (1–5)
Travelers	4	34	3.8
Shoppers	4	29	4.2
Low-tech literacy	2	58	1.5

All participants had never used Omninet AI or similar public kiosks before.

2.2 Test Tasks

The 5 core tasks described in the test plan (T1–T5).

2.3 Environment & Equipment

- Kiosk hardware: 24-inch touchscreen all-in-one running a high-fidelity interactive prototype.
- Payment simulation: test magnetic stripe card + cash simulator.
- Recording: screen capture + camera + test notes.

2.4 Metrics

- Task completion rate (%)
- Task completion time (seconds)
- Number of errors (by severity)
- SUS questionnaire (10-item Likert)
- AI Assistant satisfaction (3 items: usefulness, speed, naturalness)

3. Quantitative Results

3.1 Task Completion Rate & Average Time

Task	Completion Rate	Avg Time (sec)	Target	Met?
T1: Pay with cash and start browsing	90%	78	⩽ 60	❌
T2: AI Assistant information query	100%	28	N/A	✅
T3: Buy extra time (credit card)	100%	35	N/A	✅
T4: Language switch + AI translation	70%	112	N/A	❌
T5: End session	90%	12	N/A	✅
Overall	**88%**	—	—	—

Details of incomplete tasks:

- T1: One low-tech user inserted cash but did not know to click the "Confirm" button; gave up after timeout.
- T4: Two users (traveler + low-tech) could not find the way back to English after switching to Hebrew; one user completed switching but received an incorrect AI translation.
- T5: One user could not find the "Log out" button and simply closed the browser tab (unexpected flow).

3.2 Error Analysis

Error Type	Occurrences	Severity	Description
Forgot to click "Confirm" after cash insertion	1	Critical	Transaction incomplete, user confused

Error Type	Occurrences	Severity	Description
Lost in language selection screen	3	Critical	Could not return after switching, needed reset
AI translation incorrect or missing	2	Medium	Affected task success
Could not find “Log out” button	1	Medium	User mistakenly thought closing tab was enough
Ignored the time-extension pop-up	2	Minor	User did not notice the pop-up, session expired

3.3 Efficiency – Number of Steps

Task	Avg Touch Count	Target ⩽ 8	Met?
T1	9.2	❌	Payment flow has too many steps
T2	3.5	✅	
T3	4.1	✅	
T4	12.0	❌	Language switching path is deep
T5	2.0	✅	

3.4 SUS Score

- **Average SUS score: 76.5** (SD 9.2)
- **Range:** 62 – 88
- **Low-scoring users:** 2 low-tech users (62, 65) dissatisfied with language switching and cash payment.
- **High-scoring users:** Shoppers generally gave scores above 80.

3.5 AI Assistant Satisfaction (1–5)

Aspect	Average Score
Usefulness (did it solve the problem)	4.3
Speed (timely response)	3.5
Naturalness (conversational flow)	4.0

Users commonly complained about slow responses when the AI Assistant translated long paragraphs (> 3 seconds).

4. Qualitative Findings

From think-aloud and final interviews, the following themes emerged:

4.1 Positive Experiences

- *"The AI Assistant is quite smart – I asked for nearby coffee shops and it gave me a list."* (P03, traveler)
- *"Adding time with a credit card is fast, just like online shopping."* (P07, shopper)
- *"The interface is clean, no annoying ads."* (P01, traveler)

4.2 Main Pain Points

(1) Poor usability of language selection screen

- When users switched to a non-Latin script (e.g., Hebrew, Arabic), all interface text changed to that script, making it impossible to find the "Language" or "Settings" option.

- P09 (low-tech): *"I selected Hebrew, then I couldn't understand anything. I didn't know how to get back to English."*
- **Linked task:** T4 completion rate only 70%.

(2) Cash payment flow not intuitive enough

- After inserting cash, there was no clear prompt to click a "Confirm" button.
- Change information was displayed too small; users worried the machine would not give change.
- P10: *"I put in $10 but didn't see a confirm button. I waited for a long time thinking the machine was broken."*
- **Linked task:** T1 completion time exceeded target.

(3) AI Assistant response speed fluctuated

- Simple queries (weather, time) were fast (< 1 second), but webpage translation took 3–5 seconds.
- P04: *"I waited 4 seconds – thought it didn't respond – clicked again, and got two answers."*
- **Linked metric:** Speed satisfaction only 3.5/5.

(4) "Log out" button not prominent enough

- Some users expected a browser-style "X" close button in the corner, not the "Log out" icon on the taskbar.
- P06: *"I searched for a long time; eventually I found a little person icon in the bottom right."*

4.3 Other Observations

- Users would like voice input for the AI Assistant (currently text-only).
- Low-tech users asked for a "demo mode" or guided animation.
- Several users appreciated the privacy notice ("session data will be deleted").

5. Recommendations (Prioritised)

Priority	Issue	Suggestion	Owner
High	Cannot recover from language switch	Show language options in both native script and current interface language (already required in MRD v1.3 - ensure implementation). Add a hardware "reset language" button or voice command.	UI design, Dev
High	AI Assistant translation too slow	Optimise cloud model inference or add local caching; show loading animation when response exceeds 2 seconds to prevent double clicks.	AI team
High	Cash payment confirmation unclear	Highlight the "Confirm payment" button after cash insertion with animation; optionally add voice prompt.	Dev, UX
Medium	"Log out" button not obvious	Enlarge the "Log Out" icon and place it in the top-right corner, use red/high-contrast colour; add tooltip.	UI design
Medium	No voice input	Add microphone icon to AI Assistant for voice input, beneficial for users with limited typing ability or mobility issues.	Product, Dev
Low	Insufficient onboarding for new users	Show a short 30-second tutorial on first use (optional skip).	Product, Design

6. Conclusion

The Omninet AI kiosk demonstrates good usability for core browsing and payment functions, and users rate the AI Assistant’s usefulness highly. However, **recoverability from language switching**, **cash payment confirmation guidance**, and **AI response speed** are the main bottlenecks affecting user experience. It is recommended to address the high-priority issues before general release, and to consider voice input and onboarding animations in future iterations.

Next steps: After fixing the high-priority issues, a regression usability test will be conducted on the same tasks to verify improvements. In parallel, a usability test for the Support Center will be launched.

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